AC-15 Total Artificial Heart Emergency

MEDICAL-DIRECTOR APPROVED APP CONTENT | SOURCE-PDF REPLACEMENT | Reconciled October 16, 2026

Scope: Use for a patient with a SynCardia total artificial heart (TAH). It is not an LVAD; restore driver output rather than applying a generic cardiac-arrest algorithm.

TAH driver and perfusion pathway

1. Identify TAH	Pulsatile flow is usually present. Identify external driver, two pneumatic drivelines, power source, alarms, caregiver, and backup driver.
2. Assess circulation	Pulse/BP when obtainable, mental status, skin perfusion, breathing, ETCO2 if ventilated, and signs of shock.
3. Restore driver output	Check power, batteries, connections, kinking, alarms, and driver function. Use trained caregiver and SynCardia/implant-center support immediately.
4. Treat associated problem	Support airway/ventilation and use complaint-specific care after driver function is restored. Control bleeding and evaluate stroke, infection, or shock.
5. Hypotension/poor perfusion after driver check	Cautious NS 250-mL IV/IO boluses with reassessment only when appropriate; contact Medical Control and transplant center.
6. Transport	Bring driver, batteries/power, backup equipment, caregiver, and device information. Route to the TAH-capable/implanting center as directed.

Key treatment and handoff

Item	Approved action
High BP/pulmonary edema	Nitroglycerin 0.4 mg SL every 5 minutes only for SBP at least 150 mmHg and maintain SBP at least 90 mmHg.
Emergency support	SynCardia Clinical Emergency Support: 866-771-9437 (qualified healthcare providers).

Do not perform manual or mechanical chest compressions on a TAH patient. Do not use AED/defibrillation, cardioversion, pacing, antiarrhythmics, or routine ACLS drugs to treat TAH pump failure.

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SAFETY, LINKED PATHWAYS, AND DOCUMENTATION

Safety checks

- Do not disconnect both pneumatic drivelines simultaneously, disassemble the driver, or force a damaged connection.

- Do not confuse a TAH with an LVAD; AC-14 does not apply to TAH pump failure.

- The trained caregiver and transplant coordinator are key clinical resources and may be most familiar with the equipment.

Linked pathways

Pathway	Use when
AC-14	Use only if device is confirmed LVAD, not TAH
AR-01	Airway/ventilatory failure
UP-14	Focal neurologic deficit/stroke
AM-05	Concurrent non-device shock after output restored

Documentation and release control

- Document the presenting condition, focused findings, rhythm/ECG when applicable, all interventions with dose/route/time/response, serial vital signs, reassessments, destination decision, and notifications.

- This source PDF replaces the legacy imported sheet and matches the approved Claiborne EMS app content reconciled October 16, 2026.

- Use the linked protocol whenever the patient crosses into another clinical pathway. Follow agency release governance, training requirements, and current medication/device availability.